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First Semester M.Tech (CBCS) Degree Examinations, December 2025

25MCT112 – SENSORS & SIGNAL CONDITIONING

(For MCT)

Duration: 3 Hours] **[Max. Marks: 100**

Note: Answer **Five full** questions choosing **Two full** questions from **Module – I & Module II** & **Two full** questions from **Module – III & Module IV each** and **One full** question from **Module – V.**

			Marks	BT*	CO*	PO*
		Module – I & Module - II				
1.	a)	Define the general concepts and terminology typically used in sensor-based measurement systems, providing a brief explanation for three key terms.	10	2	1	1, 2
	b)	Explain the concept of Microsensor technology and discuss the advantages it offers over conventional sensor technologies	10	2	2	1, 2
2.	a)	Provide a comprehensive classification of sensors, detailing the various ways in which sensors can be categorized.	10	2	1	1, 2
	b)	Detail the operation of resistive gas sensors. What is the fundamental sensing mechanism that allows these sensors to detect changes in gas concentration.	10	2	2	1, 2
3.	a)	Explain the classification of sensors based on physical principles.	10	2	1	1, 2
	b)	Explain the principle and application of liquid conductivity sensors.	10	2	2	1, 2
		Module – III & Module - IV				
4.	a)	Compare and contrast capacitive and inductive sensors in terms of working principle and applications.	10	3	3	1, 2
	b)	Discuss the principles and applications of electromagnetic sensors in industrial automation.	10	3	4	1, 2
5.	a)	Explain the operation of thermal sensors and their role in temperature measurement.	10	3	3	1, 2
	b)	Explain light sensing technology and its integration in smart systems.	10	3	4	1, 2
6.	a)	Discuss the limitations of capacitive sensors in high-frequency applications.	10	3	3	1, 2
	b)	Explain the challenges in designing moisture sensors for soil applications.	10	3	4	1, 2
		Module – V				
7.	a)	Explain the concept of loading effects on sensor output and how signal conditioning mitigates these effects.	10	4	5	1, 2
	b)	Describe the types and applications of ADCs and DACs in data acquisition systems.	10	4	5	1, 2
8.	a)	Describe the ideal op-amp characteristics and their impact on circuit design.	10	4	5	1, 2
	b)	Discuss the use of passive filters in signal conditioning circuits.	10	4	5	1, 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
